

RUDRA CHAVDA

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EDUCATION

University of California, Santa Cruz <i>Bachelor of Science, Computer Engineering</i>	Expected June 2027 GPA: 3.95
– Honors: 5x Dean's List – Coursework: Data Structures and Algorithms, C Programming, Computer Architecture, Operating Systems, Software Engineering.	

EXPERIENCE

AIEA Lab (UCSC) <i>Undergraduate Researcher (Robustifying Autonomous Vehicles)</i>	Dec 2025 – Present
• Benchmarking Reinforcement Learning (RL) algorithms using Stable Baselines to evaluate autonomous vehicle safety. • Deploying the CARLA simulator in a headless environment on the Nautilus hyper-cluster (Kubernetes). • Analyzing edge-case failures by visualizing agent behaviors and reward functions post-training.	
CA-Futures (UCSC EEB & Baskin Engineering) <i>Software Developer</i>	May 2025 – Sep 2025
• Developed a full-stack conservation modeling application using Vue.js (frontend) and Py4Web (backend). • Integrated geospatial analysis tools (GDAL, Ecoscape) to track species connectivity and visualize ecological data. • Optimized simulation parameters to process large-scale biological datasets for the Ecology Lab.	
Opennote (YC S25) <i>Campus Growth Lead</i>	Jul 2025 – Present
• Spearheaded campus growth strategy, increasing user engagement by 30% and acquiring 50+ active users.	
Fassur <i>Front-End Engineering Intern</i>	June 2023 – Oct 2023
• Architected a secure, interactive web application using React.js and Tailwind CSS. • Designed a responsive front-end framework to improve data visualization for client dashboards.	

PROJECTS

Better Notes <i>TypeScript, Tailwind CSS, Vercel</i>	Spring 2025
• Engineered a full-stack note-taking platform with a custom markdown rendering engine for student study workflows. • Secured 100+ active users within 3 months of deployment to Vercel. • Live Demo: betternotes-two.vercel.app	
Div Explorer Project <i>Machine Learning, Python</i>	Fall 2025
• Developed actionable data labeling pipelines to detect duplicate discoveries and high-overlap patterns. • Implemented categorization metrics (interpretability, coverage, anomalies) to improve ML model fairness.	
Personal Portfolio <i>Next.js, Three.js, GSAP</i>	Winter 2024
• Built a high-performance portfolio site featuring 3D interactive elements using Three.js and GSAP animations.	

LEADERSHIP

Eagle Scout Project <i>Project Lead, Troop 537</i>	Summer 2024
• Directed a team of 20+ Scouts in the construction of redwood structures for Sycamore Grove Park (Livermore).	

TECHNICAL SKILLS

Languages	Python, C, C++, Java, TypeScript, JavaScript, SQL, Assembly
Frameworks	Vue.js, React, Next.js, Py4Web, Tailwind CSS, Angular
Tools	Git, Nautilus (Cluster), CARLA, Docker, GDAL, VS Code, Google Colab
Concepts	Reinforcement Learning, Data Structures, Algorithms, System Design